Frankfurt School

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Master 2025

Financial Engineering

Graded Case Studies

General Introduction

We are going to have a challenging financial market experience. Each group has to prepare 3 cases overnight and present them the next morning. We start with basic products, continue with special products, and finally turn to the "real world" approaches.

All groups must turn in their presentation in print at the start of the presenting day (9:00). Also upload the presentation in Canvas until 9:00. This is a strict rule. The presentation (power point) should be presented like a pitch in a "real world" environment to an "intelligent" client**. Every member of the group must present a part.** This applies to all types of presentation.

Two groups share a room in the FS (2.06 /2.07) for the breakout sessions. It is possible that we must change Zoom links during the week, so please refer to Canvas. It might happen that the general Zoom session is not available for the complete group work, please make sure that you have an alternative. If you have questions, please use my E-mail T.Heidorn@FS.de and I will join you via Zoom.

If possible, please work on Campus (part of the experience).

Please include on the first page the title and your group number. On the last page(s) please show names and pictures of all group members.

Due to the 24 hours access the following special rules apply:

- Students have to remain in the same area where the rooms (2.07, 2.06, Library for Bloomberg) have been booked to avoid security problems
- Students can´t move the furniture to other places (last year students moved furniture and created some problems for the Facility Team)
- Students have to bring their FS card and ID to allow the Security Teams to make controls
- Students can´t invite other persons who are not enrolled in this course to enter in the building
- From 22h the students who abandoned the building can´t enter

Case 1 (25 points):

All groups:

1a) Evaluate a Swap and a Forward Swap using the information on the following pages. (one pager (word) if you do not have the topic 1c or 1d.

1b) Prepare a 30-minute presentation of your topic below. You must turn in your presentation before the class starts on Tuesday.

Please consider especially the following three elements and present the results on one page:

- Evaluation (Pricing)
- Risk Analysis
- Key Selling Points

Topic	**Group**
1a Today's Interest Rate Situation	
1b Interest Rate Risk PV View	
1c Interest Rate Risk Flow View	
1d Interest Rate Swap / Forward Interest Rate Swap	

Case 2 (35 points):

Evaluate your product using the information on the following pages. Prepare a 30-minute presentation of your topic below.

Product	Group
2a Reverse Floater / Leveraged Floater	
2b Callable Bond	
2c Euribor in Arrears Swap	
2d Constant Maturity Swap	

Turn in the presentation by Wednesday before the class starts. Please consider especially:

- Evaluation (Pricing)
- Risk Analysis
- Key Selling Points

Case 3 (60 points):

Please suggest a portfolio for one of the following clients corresponding to today's capital market. You must turn in the 45-minute presentation on Friday (**9:00**) before the class starts.

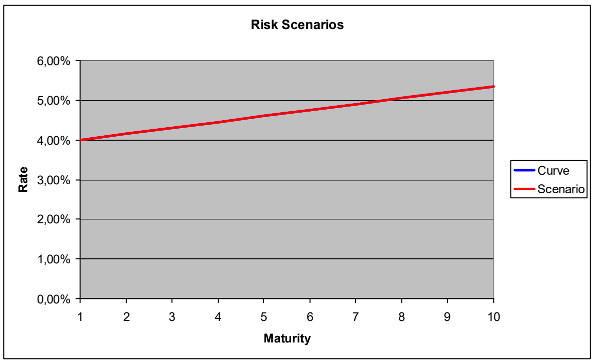
Financial Engineering Question	Group
3a Private Pension Plan	
3b Asset Manager of a Life Insurance	
3c Foreign Exchange Management for a Car Producer	
3d Kerosine Hedge for an Airline	

Market Situation for case 1 and 2:

For the pricing a single curve approach is adapted. We only use full years (no broken dates). The floating leg of a product is evaluated by using the one-year spot rate, which is identical to the one-year swap rate. All interest rates are bond base (ISDA, 30/360, exponential).

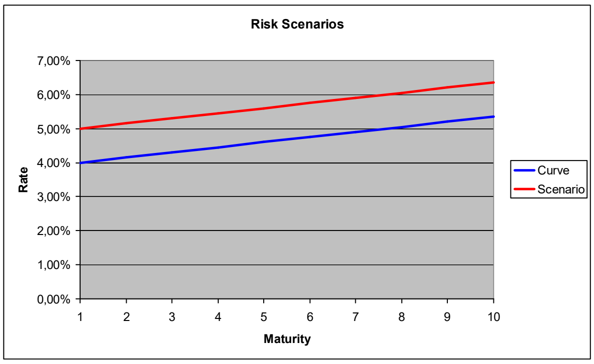
Given is the interest rate swap curve against the one-year spot rate below. The one-year rate is 4% and the slope has an incline of 0,15% per year.

Period	Swap
1	4,00%
2	4,15%
3	4,30%
4	4,45%
5	4,60%
6	4,75%
7	4,90%
8	5,05%
9	5,20%
10	5,35%



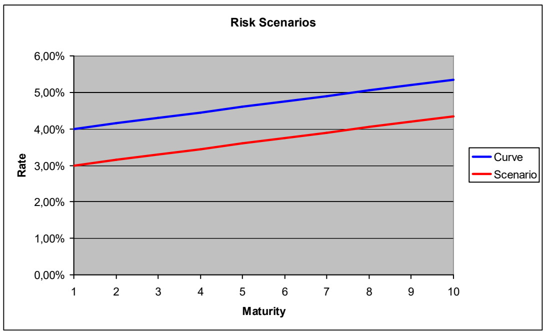
For the basic risk analysis, the following scenarios are a starting point:

1. Parallel Shift of 100 Basis Points up. (All shifts apply to the swap curve):



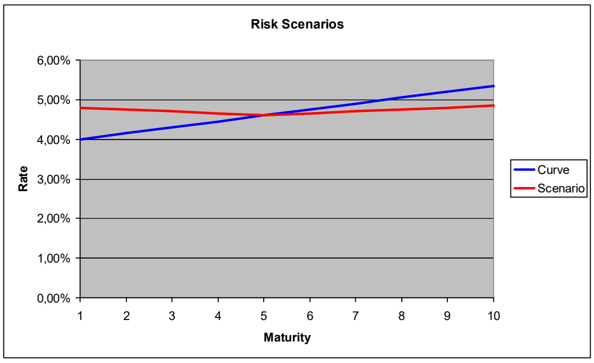
Maturity	Curve	Scenario
1	4,00%	5,00%
2	4,15%	5,15%
3	4,30%	5,30%
4	4,45%	5,45%
5	4,60%	5,60%
6	4,75%	5,75%
7	4,90%	5,90%
8	5,05%	6,05%
9	5,20%	6,20%
10	5,35%	6,35%

2. Parallel Shift of 100 Basis Points down:



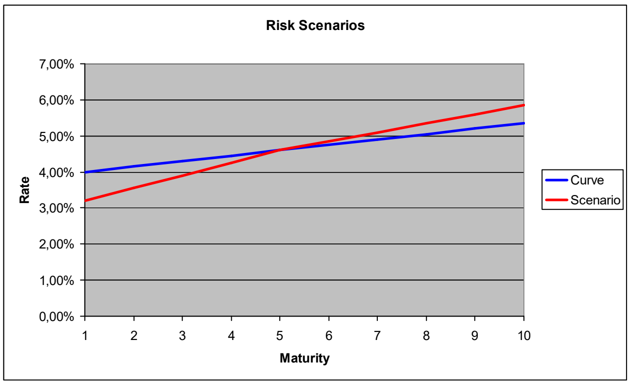
Maturity	Curve	Scenario
1	4,00%	3,00%
2	4,15%	3,15%
3	4,30%	3,30%
4	4,45%	3,45%
5	4,60%	3,60%
6	4,75%	3,75%
7	4,90%	3,90%
8	5,05%	4,05%
9	5,20%	4,20%
10	5,35%	4,35%

3. Flattening of the curve around the five-year rate which is stable
(one year rate + 80 bp, incline – 20 bp pa up to 5 years; then –10 bp pa):



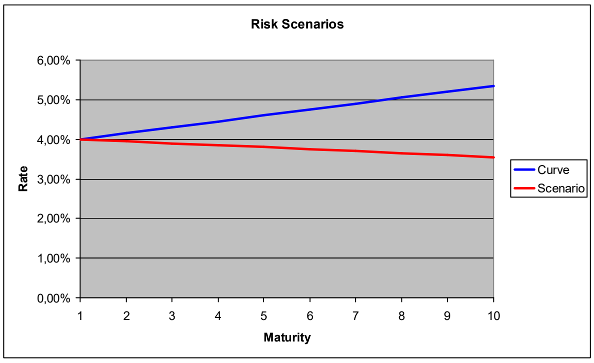
Maturity	Curve	Scenario
1	4,00%	4,80%
2	4,15%	4,75%
3	4,30%	4,70%
4	4,45%	4,65%
5	4,60%	4,60%
6	4,75%	4,65%
7	4,90%	4,70%
8	5,05%	4,75%
9	5,20%	4,80%
10	5,35%	4,85%

4. Steepening of the curve around the five-year rate which is stable
(one year rate - 80 bp, incline + 20 bp pa up to 5 years; then +10 bp pa):



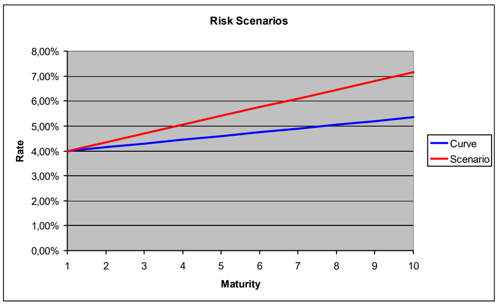
Maturity	Curve	Scenario
1	4,00%	3,20%
2	4,15%	3,55%
3	4,30%	3,90%
4	4,45%	4,25%
5	4,60%	4,60%
6	4,75%	4,85%
7	4,90%	5,10%
8	5,05%	5,35%
9	5,20%	5,60%
10	5,35%	5,85%

5. Flattening of the curve by -20 bp starting in year 2:



Maturity	Curve	Scenario
1	4,00%	4,00%
2	4,15%	3,95%
3	4,30%	3,90%
4	4,45%	3,85%
5	4,60%	3,80%
6	4,75%	3,75%
7	4,90%	3,70%
8	5,05%	3,65%
9	5,20%	3,60%
10	5,35%	3,55%

6. Steepening of the curve by +20 bp starting in year 2:



Maturity	Curve	Scenario
1	4,00%	4,00%
2	4,15%	4,35%
3	4,30%	4,70%
4	4,45%	5,05%
5	4,60%	5,40%
6	4,75%	5,75%
7	4,90%	6,10%
8	5,05%	6,45%
9	5,20%	6,80%
10	5,35%	7,15%

All groups (one pager)

Product: Interest Rate Swap

Price (**100 base**) a 5- and 10-year interest rate swap.

Risk Analysis: Receiver

Calculate the NPV changes (**100 base**) of the two swaps using the six scenarios.

Scenario	5 Years	10 Years
1 (+100)		
2 (-100)		
3 (flatter around 5)		
4 (steeper around 5)		
5 (flatter from 1)		
6 (steeper from 1)		

Product: Forward Swap

Calculate the forward swap rate in 5 for 4 and in 2 for 2.

Risk Analysis: Receiver

Calculate the NPV changes of the two swaps using the six scenarios.

Scenario	in 5 for 4 Years	in 2 for 2 Years
1 (+100)		
2 (-100)		
3 (flatter around 5)		
4 (steeper around 5)		
5 (flatter from 1)		
6 (steeper from 1)		

1 a) Group:
Today's Interest Rate Situation

Please prepare a 30-minute presentation for a corporate treasury to analyze the interest rate situation for the Euro. Show the historical development, the yield curve behavior, and explain the forecasts.

1 b) Group:
Interest Rate Risk PV View

Please prepare a 30-minute presentation on how a corporate client can assess interest rate risk using the ideas of value at risk (VAR). Show the advantages and disadvantages and discuss the calibration.

1 c) Group:
Interest Rate Risk Flow View

Please prepare a 30-minute presentation on how a corporate client can assess interest rate risk using the ideas of cash flow at risk (CAR). Show the advantages and disadvantages and discuss the calibration.

1 d) Group:
Interest Rate Swap / Forward Interest Rate Swap

Please prepare a 30-minute presentation for a corporate client explaining interest and forward interest rate swaps. Start with the single curve approach; explain the pricing and the NPV risk of the swap in question 1a. Explain the features of the product and its key characteristics for financial engineering. Show the major usage and summarize the key selling points. Also include a brief discussion on the impact of the multi-curve approach (for a corporate treasurer).

2a) Group:

Please prepare a 30-minute presentation explaining reverse and leverage floater.

Product 1: Reverse Floater

A reverse floater is a structured bond. The specific feature is the coupon; in this case the money market rate (here: one year spot) is subtracted from a fixed coupon. In our example we are using a maturity of 5 years. The Coupon is X – 12 months Euribor (= 1 year spot). The volatility of the money rate is 20%. Price the fixed rate par bond.

With the single curve approach, explain the pricing and the PV risk. Show the features of the product and its key characteristics for financial engineering. Show the major usage and summarize the key selling points.

Scenario		
1 (+100)		
2 (-100)		
3 (flatter around 5)		
4 (steeper around 5)		
5 (flatter from 1)		
6 (steeper from 1)		

Product 2: Leveraged Floater

A leveraged floater is a structured bond. The specific feature is the coupon; in this case the money market rate is doubled, and a fixed coupon is subtracted. In our example we are using a maturity of 10 years. The Coupon is twice the 12 months Euribor (= 1 year spot) minus X. The volatility of the money rate is 20%. Price the fixed rate par bond.

With the single curve approach, explain the pricing and the PV risk. Show the features of the product and its key characteristics for financial engineering. Show the major usage and summarize the key selling points.

Scenario		
1 (+100)		
2 (-100)		
3 (flatter around 5)		
4 (steeper around 5)		
5 (flatter from 1)		
6 (steeper from 1)		

2b Group:

Please prepare a 30-minute presentation to explain a single callable bond.

Product: Callable Bond

A single callable bond gives the issuer the right to redeem at a specific coupon date. For example, C5 means that the bond can be called after 5 years only.

In our example we are using a maturity of 7 years. The volatility of the interest rate is 15% for all maturities. It is not allowed to call the bond during the first two years. Find the highest fixed rate coupon for a single calling right considering a par bond.

With the single curve approach, explain the pricing and the PV risk. Show the features of the product and its key characteristics for financial engineering. Explain the major usage and summarize the key selling points. Finally, discuss the impact of multi calling rights. For example, NC2 means no call for the first two years; however, you can call the bond at every coupon date after.

Scenario		
1 (+100)		
2 (-100)		
3 (flatter around 5)		
4 (steeper around 5)		
5 (flatter from 1)		
6 (steeper from 1)		

2c Group:

Please prepare a 30-minute presentation explaining a Euribor in arrears receiver swap

Product: Euribor in Arrears Swap

We are discussing a maturity of 5 years. The product is on the fixed leg a standard swap. However, on the floating leg the fixing is adjusted. Instead of using the interest rate from the beginning of the period and paying it at the end, the interest which is fixed for the next period is used for the payment at the end of the first period. This means you pay 12-month Euribor (= 1 year spot) for the following period. The volatility is 15%.

With the single curve approach, explain the pricing and the PV risk of a receiver Euribor in arrears swap. Show the features of the product and its key characteristics for financial engineering. Explain the major usage and summarize the key selling points.

Scenario		
1 (+100)		
2 (-100)		
3 (flatter 5)		
4 (steeper 5)		
5 (flatter 1)		
6 (steeper 1)		

2d Group:

Please prepare a 30-minute presentation explaining a constant maturity swap.

Product: Constant Maturity Swap

We are analyzing a 5-year CMS paying the two-year rate and a two-year CMS paying the 5-year rate. As a base the swap structure is used. The floating leg is unchanged; in our case the client pays the 12-month Euribor (= 1-year spot rate). The other leg is the CMS leg. The CMS rate is fixed every year in advance. Therefore, the first coupon of this leg is today's 2-year swap rate (5-year swap rate). The next coupon will be the 2-year swap rate (5-year) fixed in one year. The pricing is the percentage amount which is paid on the CMS leg. The volatility is 20%.

With the single curve approach, explain the pricing and the PV risk of a Constant Maturity Swap. Explain the features of a CMS. Explain the key characteristics for financial engineering. Explain the major usage and summarize the key selling points.

Scenario		
1 (+100)		
2 (-100)		
3 (flatter 5)		
4 (steeper 5)		
5 (flatter 1)		
6 (steeper 1)		

3a Group

Please prepare a 45-minute presentation suggesting a pension plan. This time please use real world data (today), the result should be an investable portfolio.

Private Pension Plan

A professor, 45 years old, wants to invest 250 000 € as an addition to the meager German social security. He is working for a private university (Offenbach Schule der Finanzen OS). He started working with 30 after finishing his PhD. His salary development is given below. He expects that his salary will stay stable (increases only due to inflation) until his retirement.

Age	**45**	**40**	**35**	**30**
Income €/Year	81.600	72.000	48.000	36.000

He wants to start the pension at 68 and has no other private pension plan. He owns a house, is married (his wife is 43) and has two children (5 and 8). His wife is not working and is not planning to do so. He probably does not need any liquidity from this investment until the pension starts. His risk attitude is slightly risk averse, especially considering pension plans. The family has a private health insurance (now 700 €/month).

He is a finance professor, even though he teaches in Offenbach, he has a very good understanding of financial markets.

3b Group

Please prepare a 45-minute presentation suggesting an investment plan for the asset management of a life insurance. This time please use real world data (today), the result should be an investable portfolio.

Asset Manager of a Life Insurance

The All-Kanns life insurance is looking for a partner to improve the investment process. Its manager has an MA in Finance from a school in a small village in Germany (Offenbach see case 3a). The portfolio consists of 100 000 insured persons. All persons are 50 years old (50% male) and so far the portfolio value per person is 50 000,-- €. However, for the next ten years each person will pay an additional 5 000,-- € per year (at the end of each year, starting this year).

Five years after their last payment the insurants have the choice of either taking a lump sum payment or a monthly pension. The guarantied interest rate is 1,0% (Höchstrechnungszins). We assume that nobody dies during the first 15 years. If the person chooses the pension plan, the pension stops in the year he/she dies. However, the pension is paid for at least 10 years. Please develop an investment strategy. The accounting principle is IAS.

3c Group

Please prepare a 45-minute presentation to present a suggestion for foreign currency management to a German car producer. This time please use real world data (today), the result should be investable.

Foreign Exchange Management for a Car Producer

A small German sports car factory (Bursche) was not bought by VW. They have a single plant in Germany. Because of the relatively small production it is not feasible to open plants in other countries.

The firm wants to restructure its foreign exchange management. Please develop a proposal for the next five years. As a base scenario Bursche has given you the following data.

The operating expenses are given below. They are in Euro only and this is not changing for the next four years. Also given below are operational revenues. They are converted by using the currencies mix given today for T. However, you only have to analyze the FX risk forward looking (from T+1) using T (= now) as a base year. The firm expects stable production at last year's level (data at T). Starting with the currency mix in T the sales will further shift to the Asian region (China see the forecast below). The selling price now is € 100.000 in all countries. However, the prices in local currency cannot be changed in the future. The balance sheet volume is 28.000 Mio Euro. This is financed with 40% euro denominated equity and 60% with euro denominated liabilities. Last year's profit was 1 000 Mio Euros.

		T	T+1	T+2	T+3	T+4
Total Operating Revenue (Mio EUR)		7.530				
Total Operating Expenses (Mio EUR)		5455				
Total Operating Income (Mio EUR)		2075				
Operatinng Income Currency Mix						
	EUR	45%	41%	37%	35%	33%
	US$	44%	44%	44%	44%	44%
	RMB=Renminbi	11%	15%	19%	21%	23%

3d Group

Please prepare a 45-minute presentation suggesting a kerosene hedge for a US airline. This time please use real world data (today), the result should be based on investable products.

Kerosene Hedge for a US Airline

A US airline (Lusthansa) is operating on US$ base only. The firm wants to restructure its kerosene price risk management. Please develop a proposal for the next two years. As a base scenario Lusthansa has given you the following projection of the estimated need for kerosene for the next two years. There is no hedging in place for that period.

Due to the fierce competition in the American market ticket prices are difficult to adjust. A substantial part of the expenses is fuel cost, therefore Lusthansa wants to hedge. However, there is some uncertainty about the need of kerosene, due to the development of the market.

	Year					
	Q1/25	Q2/25	Q3/25	Q4/25	H1/26	H2/26
Volume kerosene m. tons	2000	2500	2500	3000	6000	7000